

15. Experiment: AI-based Traffic Prediction for 5G Networks

Objective1:

Code:

```
clc;
```

```
clear;
```

```
close all;
```

```
% Simulated 5G Network Traffic (Mbps)
```

```
time = 1:20;
```

```
traffic = [20 25 30 35 40 50 60 58 65 70 ...  
          75 80 78 85 90 95 100 98 105 110];
```

```
figure;
```

```
plot(time, traffic, '-o', 'LineWidth', 2);
```

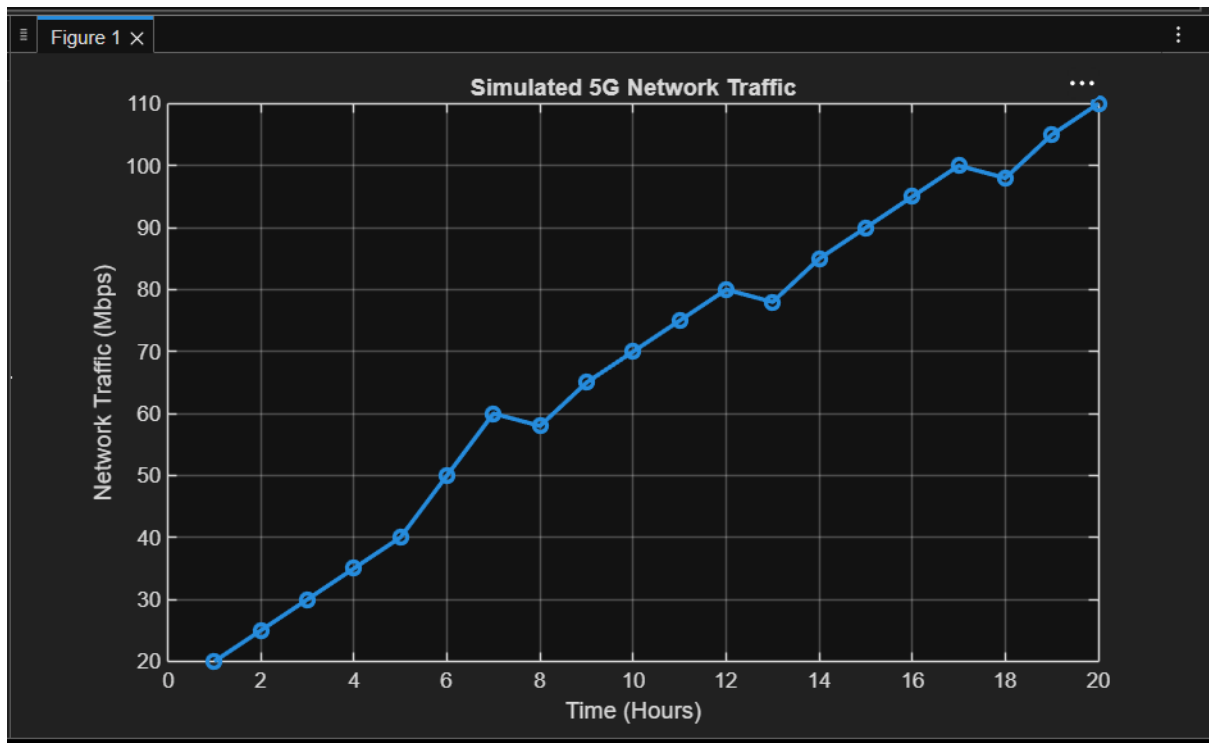
```
grid on;
```

```
xlabel('Time (Hours)');
```

```
ylabel('Network Traffic (Mbps)');
```

```
title('Simulated 5G Network Traffic');
```

Output:



Objective 2:

Code: clc;

clear;

close all;

% Simulated Traffic Data

time = 1:20;

traffic = [20 25 30 35 40 50 60 58 65 70 ...
75 80 78 85 90 95 100 98 105 110];

% Linear Regression Model

model = polyfit(time, traffic, 1);

% Predicted Traffic

predicted = polyval(model, time);

% Plot Actual and Predicted Traffic

```
figure;
```

```
plot(time, traffic, 'bo-', 'LineWidth', 2);
```

```
hold on;
```

```
plot(time, predicted, 'r*-', 'LineWidth', 2);
```

```
grid on;
```

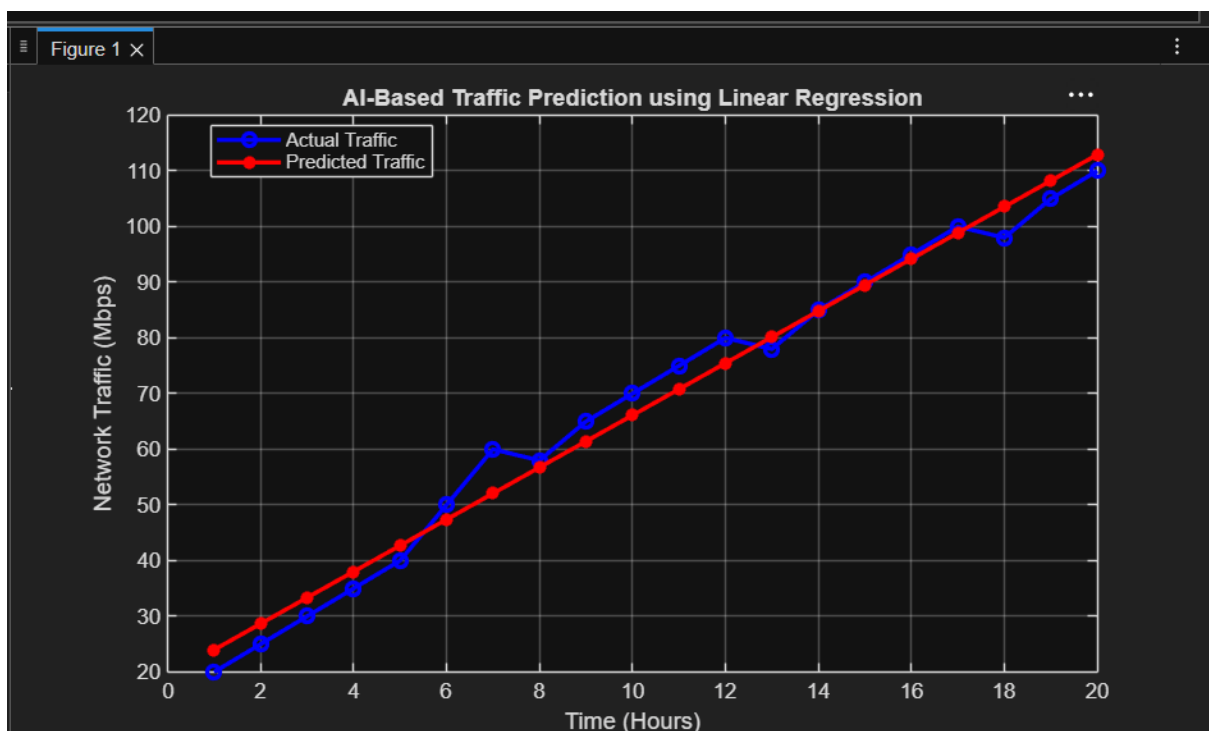
```
xlabel('Time (Hours)');
```

```
ylabel('Network Traffic (Mbps)');
```

```
title('AI-Based Traffic Prediction using Linear Regression');
```

```
legend('Actual Traffic', 'Predicted Traffic', 'Location', 'best');
```

Output:



Objective 3:

Code:

```
clc;

clear;

close all;

% Simulated Traffic Data
time = 1:20;

traffic = [20 25 30 35 40 50 60 58 65 70 ...
          75 80 78 85 90 95 100 98 105 110];

% Linear Regression Model
model = polyfit(time, traffic, 1);

% Predicted Traffic
predicted = polyval(model, time);

% Calculate Prediction Error
error = traffic - predicted;

% Calculate RMSE
RMSE = sqrt(mean(error.^2));

disp(['Root Mean Square Error (RMSE) = ', num2str(RMSE)]);

% Plot Results
figure;

% Actual vs Predicted Traffic
subplot(2,1,1);
plot(time, traffic, 'bo-', 'LineWidth', 2);
hold on;
```

```
plot(time, predicted, 'r*-', 'LineWidth', 2);
```

```
grid on;
```

```
xlabel('Time (Hours)');
```

```
ylabel('Traffic (Mbps)');
```

```
title('Actual vs Predicted Traffic');
```

```
legend('Actual', 'Predicted', 'Location', 'best');
```

```
% Prediction Error
```

```
subplot(2,1,2);
```

```
bar(time, error);
```

```
grid on;
```

```
xlabel('Time (Hours)');
```

```
ylabel('Prediction Error (Mbps)');
```

```
title('Prediction Error');
```

```
% Display Results
```

```
disp('Time  Actual  Predicted');
```

```
disp([time' traffic' predicted']);
```

```
output:
```

